

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Artificial Intelligence and Data Science**  
**ASSESSMENT TOOL 1 – Experiments**

|                         |                                                                                                                                |                      |                                 |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------|
| <b>Course Code:</b>     | DSA03                                                                                                                          | <b>Course Title:</b> | Natural Language Processing     |
| <b>CO Assessed:</b>     | CO3 – Precision                                                                                                                | <b>Assessment:</b>   | Assessment Tool 3 – Experiments |
| <b>Weightage:</b>       | 50% of CIA                                                                                                                     |                      |                                 |
| <b>CO5 Statement:</b>   | Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3) |                      |                                 |
| <b>Student Name:</b>    | Harshavardhan. RV                                                                                                              | <b>Reg. No.:</b>     | 192472163                       |
| <b>Section / Batch:</b> | CSE-AI                                                                                                                         | <b>Date:</b>         | 8/8/2026                        |

**Code of Conduct**

*I \_\_\_\_ Harshavardhan RV(192472163)\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: \_\_\_\_ HARSHAVARDHAN. RV\_\_\_\_\_

**Instructions**

- Perform all experiments using Google Colab.
- Create a separate notebook (.ipynb) for the assessment.
- Ensure all code is executable without errors.
- Include appropriate comments explaining the logic used.
- Complete all three experiments in a single Google Colab notebook.
- Execute all cells and display the outputs for the given test cases.
- Save the notebook with the naming convention: RegNo\_Name\_CO3-AT1.ipynb
- Upload the notebook to your GitHub repository.
- Ensure the repository is public or accessible to the instructor.
- Submit the following:
  - GitHub Repository Link in GCR
  - Google Colab Share Link in GCR
  - Screenshots of uploaded coding and output files as printout.
- Duration: 30 minutes.

| Section / Topic                                                                                                                                 | Focus                                                                                                                                               | Experiment Questions |
|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Context-Free Grammars for English Syntax, Context-Free Rules and Trees, Sentence-Level Constructions, Parsing, Top-Down Parsing, Earley Parsing | Design a Python program to validate sentences using CFG production rules and construct parse tree representations for valid sentences.              | Q1                   |
| Agreement, Feature Structures, Sub Categorization                                                                                               | Design a Python program to verify subject-verb agreement and validate grammatical constraints using feature structures and subcategorization rules. | Q2                   |
| Probabilistic Context-Free Grammars (PCFGs)                                                                                                     | Design a Python program to parse sentences and compute sentence probabilities using probabilistic grammar production rules.                         | Q3                   |

# 50 MCQs – NLP

## Correct Answers

*Questions with all options and the correct option clearly marked*

### UNIT I – Introduction and Morphology

#### 1. What is the primary goal of Natural Language Processing (NLP)?

- A) To increase computer hardware speed
- B) To enable computers to process and understand human language
- C) To design operating systems
- D) To manage computer networks

Correct Answer: (B) To enable computers to process and understand human language

#### 2. Which model is commonly used to describe whether a string belongs to a particular language?

- A) Database model
- B) Mathematical model
- C) Financial model
- D) Hardware model

Correct Answer: (B) Mathematical model

#### 3. Which symbol is commonly used in regular expressions to represent zero or more occurrences of the preceding element?

- A) +
- B) ?
- C) \*
- D) ^

Correct Answer: (C) \*

#### 4. Which regular expression represents a sequence containing one or more occurrences of a?

- A) a\*
- B) a+
- C) a?
- D) a|

Correct Answer: (B) a+

#### 5. What is the main function of a Finite State Automaton (FSA)?

- A) To store large databases
- B) To recognize patterns in strings
- C) To translate programming languages

D) To generate images

Correct Answer: (B) To recognize patterns in strings

**6. Which type of morphology deals with grammatical variations such as tense and number?**

A) Derivational morphology

B) Inflectional morphology

C) Computational morphology

D) Structural morphology

Correct Answer: (B) Inflectional morphology

**7. Which example represents derivational morphology?**

A) cat → cats

B) walk → walked

C) happy → happiness

D) book → books

Correct Answer: (C) happy → happiness

**8. What is the purpose of finite-state morphological parsing?**

A) To identify the internal morphological structure of words

B) To identify network protocols

C) To calculate sentence length

D) To generate numerical data

Correct Answer: (A) To identify the internal morphological structure of words

**9. What does the Porter Stemmer primarily perform?**

A) Sentence parsing

B) Word stemming

C) POS tagging

D) Speech recognition

Correct Answer: (B) Word stemming

**10. The word “unhappiness” can be analyzed as un + happy + ness. Which process is illustrated by this analysis?**

A) Inflectional morphology only

B) Derivational morphology

C) Syntax parsing

D) Regular expression matching

Correct Answer: (B) Derivational morphology

## UNIT II – Word Level and Syntactic Analysis

### 11. What does an N-gram model estimate?

- A) Probability of a word based on previous words
- B) Number of letters in a word
- C) Meaning of a paragraph
- D) Number of sentences in a document

Correct Answer: (A) Probability of a word based on previous words

### 12. A bigram model considers how many words at a time?

- A) One
- B) Two
- C) Three
- D) Four

Correct Answer: (B) Two

### 13. What is the main problem with an unsmoothed N-gram model?

- A) It cannot count words
- B) It assigns zero probability to unseen N-grams
- C) It cannot recognize nouns
- D) It cannot process sentences

Correct Answer: (B) It assigns zero probability to unseen N-grams

### 14. What is the primary purpose of smoothing in language models?

- A) To remove punctuation
- B) To assign probabilities to unseen events
- C) To identify verbs
- D) To shorten sentences

Correct Answer: (B) To assign probabilities to unseen events

### 15. What does entropy measure in an N-gram language model?

- A) Linguistic uncertainty
- B) Number of nouns
- C) Sentence length
- D) Number of characters

Correct Answer: (A) Linguistic uncertainty

### 16. Which word class does “quickly” belong to in the sentence “She runs quickly”?

- A) Noun
- B) Verb
- C) Adjective

D) Adverb

Correct Answer: (D) Adverb

**17. What is the purpose of a tagset in POS tagging?**

- A) To define the set of grammatical labels used for words
- B) To store sentences
- C) To generate regular expressions
- D) To calculate entropy

Correct Answer: (A) To define the set of grammatical labels used for words

**18. In the sentence “Book a ticket,” what is the POS of “Book”?**

- A) Noun
- B) Verb
- C) Adjective
- D) Adverb

Correct Answer: (B) Verb

**19. Which POS tagging approach uses manually written linguistic rules?**

- A) Stochastic tagging
- B) Rule-based tagging
- C) Transformation-based tagging
- D) Random tagging

Correct Answer: (B) Rule-based tagging

**20. Which POS tagging method combines an initial tagging with a sequence of corrective transformations?**

- A) Rule-based tagging
- B) Stochastic tagging
- C) Transformation-based tagging
- D) Dictionary-based tagging

Correct Answer: © Transformation-based tagging

## UNIT III – Context-Free Grammars

**21. What is the primary purpose of a Context-Free Grammar (CFG) in NLP?**

- A) To represent syntactic structure of sentences
- B) To calculate word frequency
- C) To identify word meanings
- D) To translate documents

Correct Answer: (A) To represent syntactic structure of sentences

**22. Which component of a CFG specifies how a non-terminal can be expanded?**

- A) Terminal symbol
- B) Production rule
- C) Probability table
- D) Lexicon only

Correct Answer: (B) Production rule

**23. In the rule  $S \rightarrow NP VP$ , what does S usually represent?**

- A) Sentence
- B) Subject
- C) Semantics
- D) String

Correct Answer: (A) Sentence

**24. Which structure visually represents the syntactic organization of a sentence?**

- A) Bar chart
- B) Parse tree
- C) Flowchart
- D) Frequency table

Correct Answer: (B) Parse tree

**25. What does grammatical agreement ensure?**

- A) Words have the same number of letters
- B) Related words have compatible grammatical features
- C) All words have the same meaning
- D) Sentences contain equal numbers of words

Correct Answer: (B) Related words have compatible grammatical features

**26. What does subcategorization describe?**

- A) The arguments or complements that a word, especially a verb, requires
- B) The number of letters in a word
- C) The probability of a sentence

D) The meaning of punctuation

Correct Answer: (A) The arguments or complements that a word, especially a verb, requires

**27. What is parsing in NLP?**

- A) Converting text into numbers only
- B) Determining the syntactic structure of a sentence
- C) Removing stop words
- D) Translating words between languages

Correct Answer: (B) Determining the syntactic structure of a sentence

**28. Which parsing strategy begins with the start symbol and attempts to generate the input sentence?**

- A) Bottom-up parsing
- B) Top-down parsing
- C) Stochastic tagging
- D) Semantic parsing

Correct Answer: (B) Top-down parsing

**29. Which parsing algorithm can parse sentences using dynamic programming and handle left-recursive grammars?**

- A) Porter algorithm
- B) Earley parser
- C) Viterbi tagger
- D) PageRank algorithm

Correct Answer: (B) Earley parser

**30. What is the purpose of a Probabilistic Context-Free Grammar (PCFG)?**

- A) To associate probabilities with grammar productions
- B) To remove grammar rules
- C) To identify word stems
- D) To classify documents

Correct Answer: (A) To associate probabilities with grammar productions

## UNIT IV – Semantic Analysis

### 31. What is the main objective of semantic analysis?

- A) To determine the meaning of linguistic expressions
- B) To count characters
- C) To identify punctuation
- D) To format documents

Correct Answer: (A) To determine the meaning of linguistic expressions

### 32. Which formal system is commonly used to represent relationships between entities and properties?

- A) First-Order Predicate Calculus
- B) Regular Expression
- C) Finite Automaton
- D) N-gram model

Correct Answer: (A) First-Order Predicate Calculus

### 33. In first-order predicate logic, what does $P(x)$ typically represent?

- A) A probability
- B) A predicate applied to an argument
- C) A parsing rule
- D) A POS tag

Correct Answer: (B) A predicate applied to an argument

### 34. Which representation can express the statement “John loves Mary” as a relationship between entities?

- A) loves(John, Mary)
- B) John + Mary
- C) John/Mary
- D) John = Mary

Correct Answer: (A) loves(John, Mary)

### 35. What is syntax-driven semantic analysis?

- A) Deriving meaning representations based on syntactic structure
- B) Counting syntactic words
- C) Removing grammatical rules
- D) Generating random meanings

Correct Answer: (A) Deriving meaning representations based on syntactic structure

### 36. What is a lexeme?

- A) An abstract vocabulary unit representing related word forms

- B) A punctuation mark
- C) A sentence structure
- D) A grammar rule

Correct Answer: (A) An abstract vocabulary unit representing related word forms

**37. What does word sense refer to?**

- A) The grammatical category of a word
- B) A particular meaning of a word
- C) The number of letters in a word
- D) The pronunciation of a word

Correct Answer: (B) A particular meaning of a word

**38. What is the main goal of Word Sense Disambiguation (WSD)?**

- A) To determine the intended meaning of an ambiguous word
- B) To identify sentence boundaries
- C) To count words
- D) To remove prefixes

Correct Answer: (A) To determine the intended meaning of an ambiguous word

**39. In “The fisherman sat on the bank,” which meaning of “bank” is most appropriate?**

- A) Financial institution
- B) River side
- C) Storage facility
- D) Computer memory

Correct Answer: (B) River side

**40. What is the primary goal of Information Retrieval (IR)?**

- A) To retrieve relevant information from a collection of documents
- B) To generate grammar rules
- C) To identify speech sounds
- D) To stem every word

Correct Answer: (A) To retrieve relevant information from a collection of documents

## UNIT V – Language Generation and Discourse Analysis

### 41. What does discourse analysis primarily study?

- A) Language beyond individual sentences and its context
- B) Individual characters only
- C) Computer hardware
- D) Word spelling only

Correct Answer: (A) Language beyond individual sentences and its context

### 42. What is reference resolution?

- A) Determining what an expression such as “he” or “it” refers to
- B) Finding the number of words in a sentence
- C) Identifying punctuation marks
- D) Translating a sentence

Correct Answer: (A) Determining what an expression such as “he” or “it” refers to

### 43. In the sentence “Ravi bought a car. He likes it,” what does “He” refer to?

- A) Car
- B) Ravi
- C) Likes
- D) It

Correct Answer: (B) Ravi

### 44. What does text coherence refer to?

- A) The meaningful and logical relationship among parts of a text
- B) The number of words in a paragraph
- C) The spelling of words
- D) The number of punctuation marks

Correct Answer: (A) The meaningful and logical relationship among parts of a text

### 45. What is a discourse act or dialogue act?

- A) The communicative function performed by an utterance
- B) The number of words in dialogue
- C) A grammatical error
- D) A punctuation rule

Correct Answer: (A) The communicative function performed by an utterance

### 46. Which of the following is an example of a dialogue act?

- A) Request
- B) Noun
- C) Prefix

D) Stem

Correct Answer: (A) Request

**47. What is the main purpose of a conversational agent?**

- A) To interact with users through natural language
- B) To perform only mathematical calculations
- C) To store images
- D) To compile source code

Correct Answer: (A) To interact with users through natural language

**48. What is the purpose of surface realization in language generation?**

- A) To convert an abstract representation into natural-language text
- B) To identify word senses
- C) To parse a sentence
- D) To calculate N-gram entropy

Correct Answer: (A) To convert an abstract representation into natural-language text

**49. Which machine translation approach uses an intermediate language-independent representation?**

- A) Direct translation
- B) Interlingua
- C) Word-for-word translation
- D) Rule deletion

Correct Answer: (B) Interlingua

**50. What do statistical approaches to machine translation primarily learn from?**

- A) Parallel and bilingual language data
- B) Only punctuation marks
- C) Computer hardware specifications
- D) Regular expressions alone

Correct Answer: (A) Parallel and bilingual language data

## Rubrics for Calculation

| Criteria                  | Weightage | Level 4 – Exemplary                                                                           | Level 3 – Proficient                                        | Level 2 – Developing                                | Level 1 – Beginning                              |
|---------------------------|-----------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------|--------------------------------------------------|
| Understanding of Concepts | 25        | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design       | 30        | Well-planned, systematic implementation with optimal solution                                 | Correct implementation with minor issues                    | Basic implementation with errors or inefficiencies  | Incorrect or incomplete implementation           |
| Use of Tools              | 20        | Effective and advanced use of tools/technologies with proper configuration                    | Appropriate use of tools with correct execution             | Limited or inconsistent use of tools                | Incorrect or minimal use of tools                |
| Analysis of Results       | 15        | Insightful analysis with accurate interpretation and validation of results                    | Good analysis with reasonable interpretation                | Basic analysis with limited insights                | Poor or incorrect analysis of results            |
| Documentation             | 10        | Clear, well-structured documentation with diagrams, code, and results                         | Organized documentation with necessary details              | Basic documentation with missing details            | Poor or incomplete documentation                 |

| Q. No. / Criteria Level | Understanding of Concepts | Experimental Design | Use of Tools | Analysis of Results | Documentation | Sub. Total | Total % |
|-------------------------|---------------------------|---------------------|--------------|---------------------|---------------|------------|---------|
| 1                       |                           |                     |              |                     |               |            |         |
| 2                       |                           |                     |              |                     |               |            |         |
| 3                       |                           |                     |              |                     |               |            |         |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |